

Update on 2025 and 2026 tracking efficiencies

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2nd June 2026

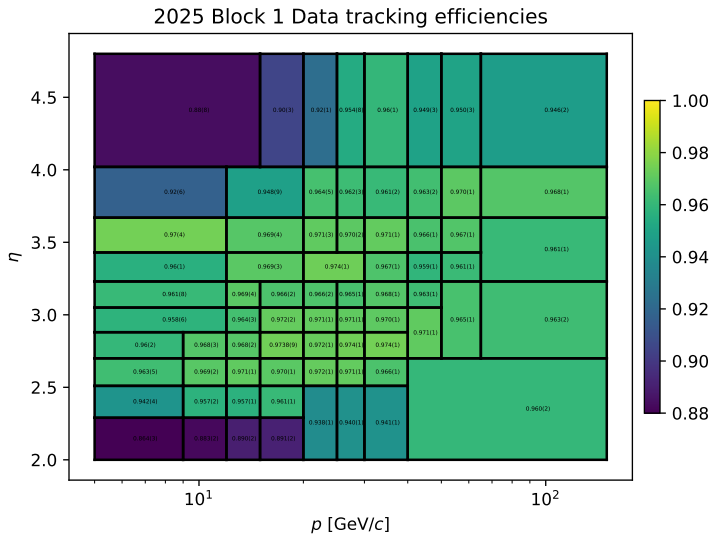


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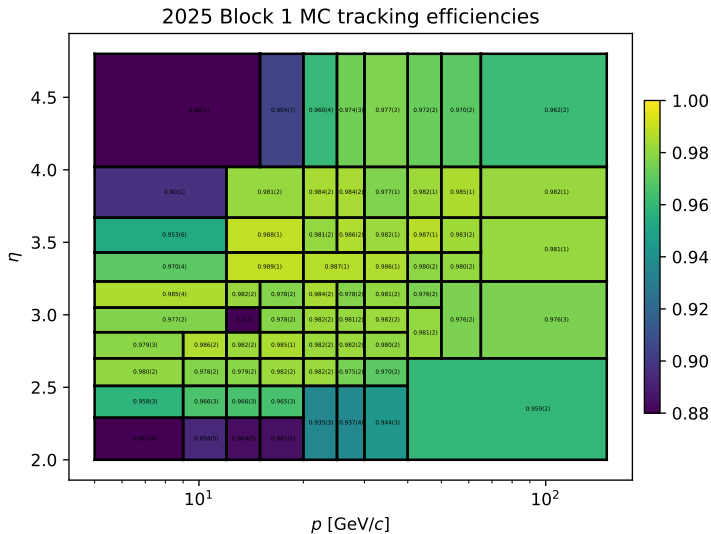
Progress update:

- ① Tracking efficiency tables produced for 2025 block 1 and 2 (pre-TS)
 - Velo M23 turned off in MC, but not in data
- ② Tracking efficiencies fitted to all of 2025 and 2026 data using 2025 pre-TS MC to constrain fit shapes
 - Radiation damage in the central region visible
 - Damaged Velo clearly visible in highest η bin and high momentum
 - Runs with muon hole checked
- ③ Plan to present in RTA WP4 on Thursday
- ④ MC request for 2025 post-TS approved, delayed because MC liason was away

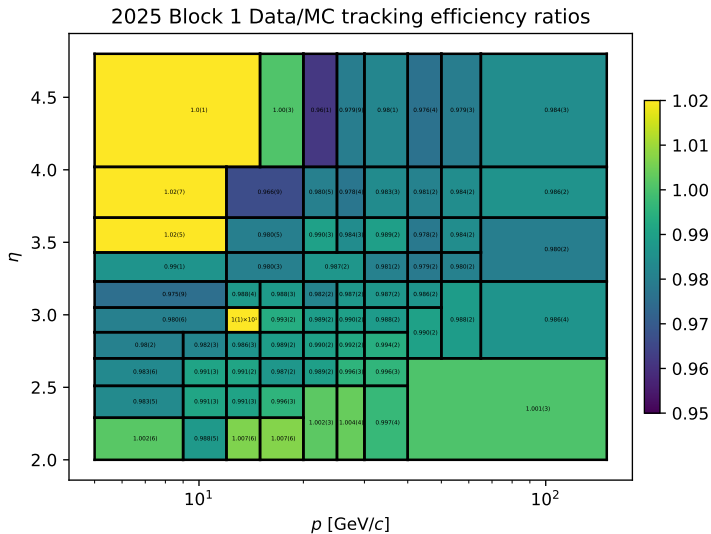
2025 pre-TS tracking efficiency corrections



2025 pre-TS tracking efficiency corrections

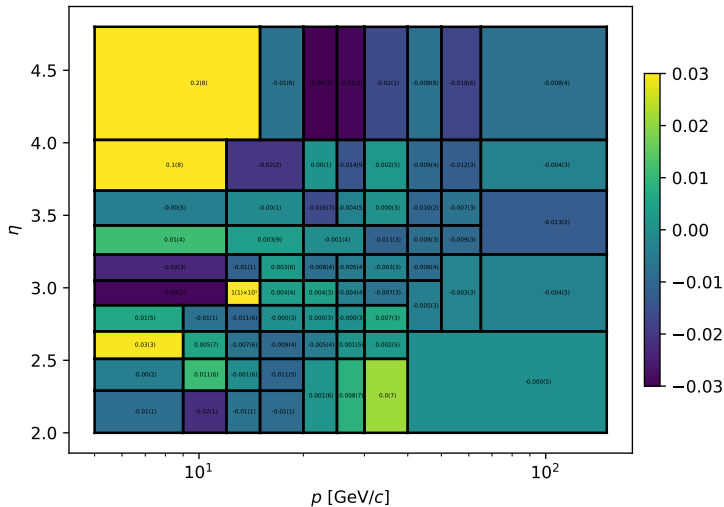


2025 pre-TS tracking efficiency corrections

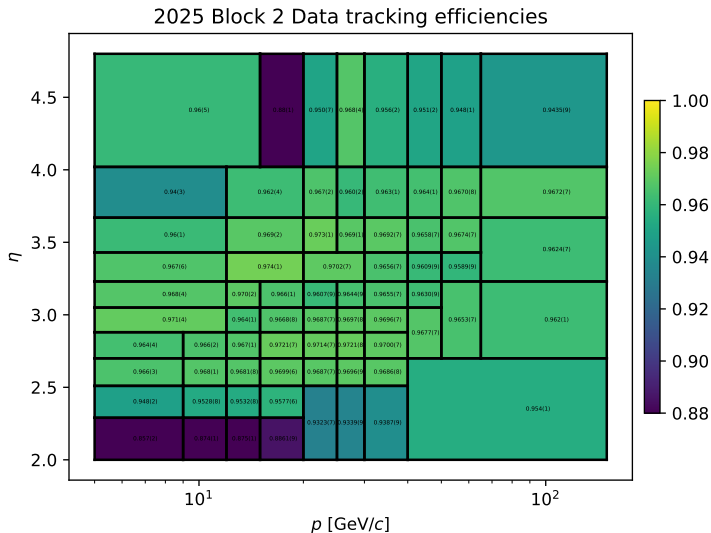


2025 pre-TS tracking efficiency corrections

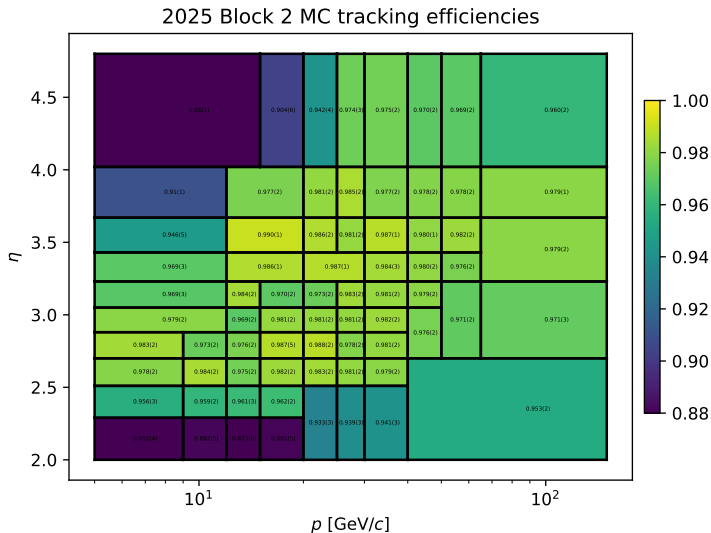
2025 Block 1 Data/MC Combined vs MuonUT ratio difference



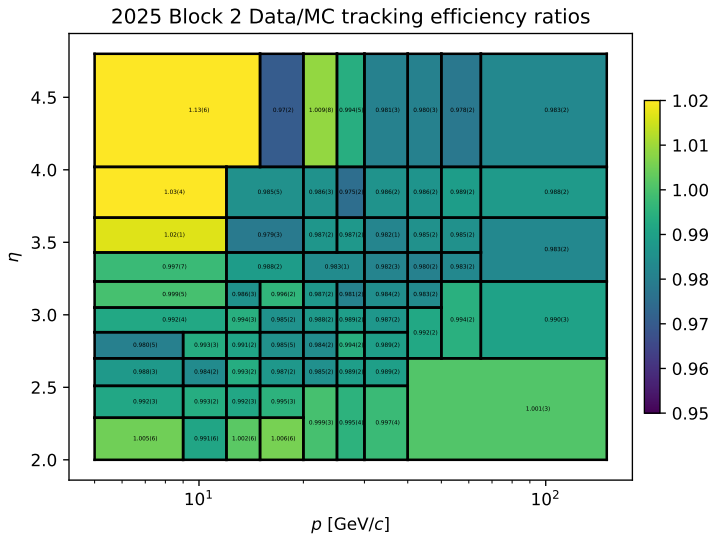
2025 pre-TS tracking efficiency corrections



2025 pre-TS tracking efficiency corrections

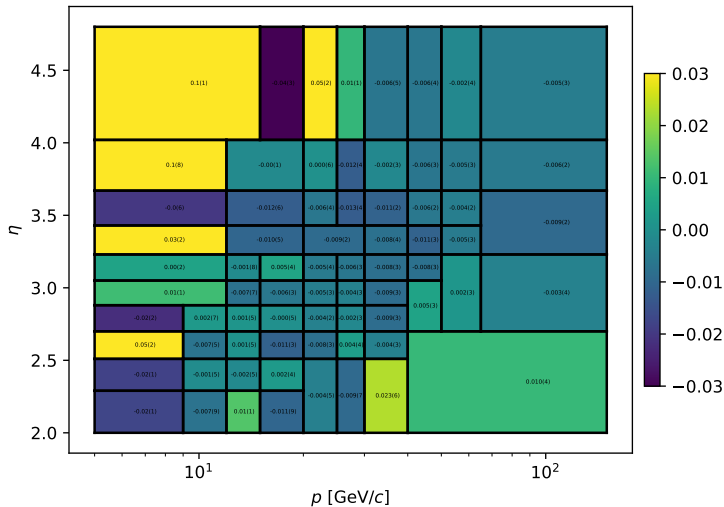


2025 pre-TS tracking efficiency corrections



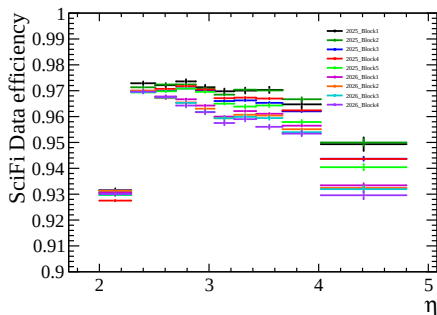
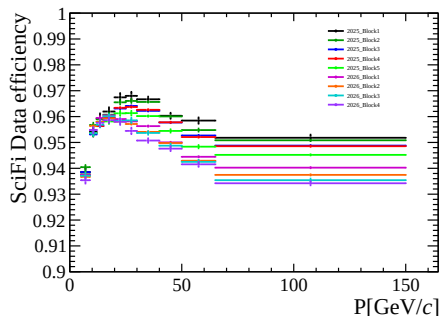
2025 pre-TS tracking efficiency corrections

2025 Block 2 Data/MC Combined vs MuonUT ratio difference



2025 and 2026 data tracking efficiencies

Comparison of tracking efficiencies in 2025 and 2026 data blocks

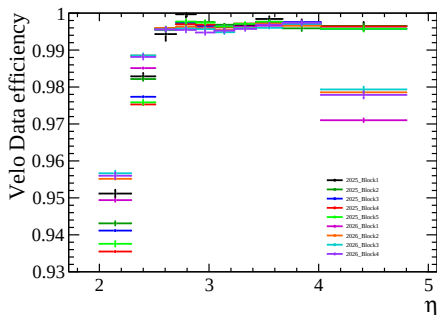
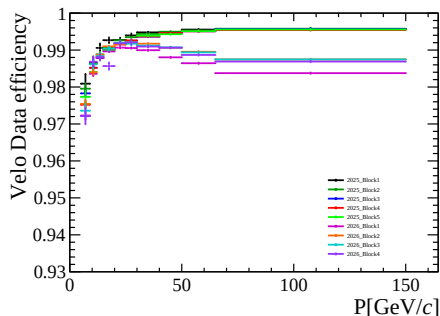


VeloMuon track efficiencies

Looks like radiation damage in the central region

2025 and 2026 data tracking efficiencies

Comparison of tracking efficiencies in 2025 and 2026 data blocks

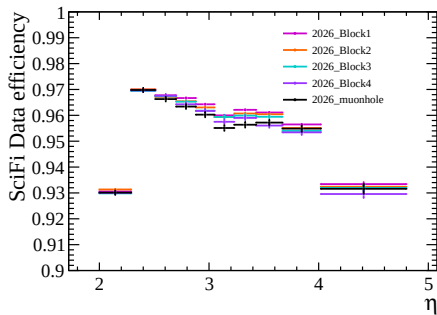
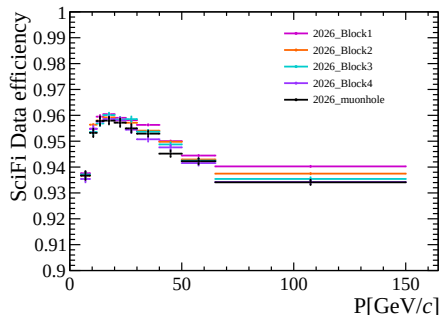


Downstream track efficiencies

Velo damage at the end of 2025 clearly seen!

Muon hole in 2026 data tracking efficiencies

Check tracking efficiencies in CONDITIONAL runs with muon hole
([see here](#))

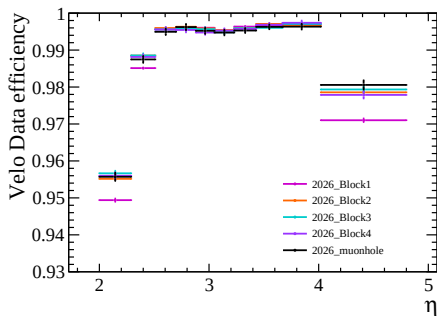
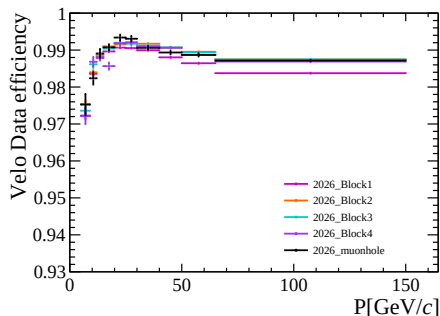


VeloMuon track efficiencies

Maybe small drop near $\eta \approx 3.5$

Muon hole in 2026 data tracking efficiencies

Check tracking efficiencies in CONDITIONAL runs with muon hole
([see here](#))



Downstream track efficiencies
No effect seen!

Binning in Velo halves

Issue raised last time:

- We need to bin the two Velo halves separately...
- ...but which variable to use?
- PHI is most convenient for analysts
- NVPHITSA/C functors gives the number of hits in each Velo half
- I did a quick quantitative study: When binning in PHI, around 15%–18% of tracks are assigned the wrong Velo half
- NVPHITSA/C can also be ambiguous, happens for 3% of the tracks in VeloMuon
- But in Downstream, Velo hits are not defined for the probe track, and I don't think it makes sense to use the Velo hits of the best matched track if it's not matched...?

Not much to do for now:

- ① Wait for large 2024 MC production
- ② Wait for 2025 post-TS MC